

11.8 A 5A 94.5% Peak Efficiency 9~16V-to-1V Dual-Path Series-Capacitor Converter with Full Duty Range and Low V-A Metric

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As the power demand of computing cores grows explosively, large conversion ratio, high efficiency and compact 12V-to-1V DC-DC converters become increasingly important. Double step-down (DSD) converters [1–2], also known as series-capacitor converters, offer a promising solution using the capacitors to reduce the high voltage stress and to extend the duty cycles, however, at the cost of multiple inductors and limited duty cycle range ($D < 50\%$). To break the duty cycle limit, some previous work employs extra power switches or capacitor cross-connected (CCC) design for $D > 50\%$ operation during a short transient period [3–4]. The steady-state operation, however, is still not allowed and limits the transient response and output voltage range. Besides, DSD converters suffer from limited efficiency since the switch voltage stress is still high. To quantify the switch losses of the DSD and the proposed work, the popular V-A metric is adopted here [5], which identifies the overall converter conduction and switching losses with the lower V-A value indicating lower losses. In this work, we propose a dual-path series-capacitor converter which shows inherent full duty cycle range, low V-A metric and excellent efficiency even with only one inductor. Compared with the conventional DSD converters, the capacitors in this work not only reduces the switch voltage stress, but also form a capacitive current path. Therefore, both the switch voltage and the current stresses are alleviated, leading to competitive converter V-A metric and efficiency over wide output voltage and load ranges. In addition, the capacitor charge balance is easily achieved and, hence, inherent full duty cycle operation is obtained.

Figure 11.8.1 demonstrates the detailed power stage of the proposed dual-path series-capacitor converter which consists of the switches M_{H1} – M_{H7} and M_{L1} – M_{L2} , the flying capacitors C_{H1} – C_{L2} , and the inductor L. Similar to other series-capacitor converters, the capacitors reduce the input voltage stress during the inductor magnetization period. While during the inductor demagnetization period, the capacitors constitute a capacitive current path through the switch M_{H6} directly to the output. Hence, the inductor current stress is alleviated and duty cycle beyond 50% is naturally supported. Besides, since the voltage stress is low, all the power switches are implemented with on-chip 1.8V (M_{L2} and M_{H6}) or 5V devices. According to the V-A metric summary shown on Fig. 11.8.1 (top), the proposed work successfully reduces the switch voltage and current stresses and achieves low V-A metric, indicating an excellent power efficiency. Figure 11.8.1 also gives the output voltage and normalized inductor current versus duty cycles, which shows $D > 50\%$ operation and inductor current reduction as D increases. In addition, as will be shown later, the proposed design is compatible with most existing control schemes without exploiting complex phase shift or capacitor voltage balance techniques, as other series-capacitor converters usually have to.

Figure 11.8.2 (top) illustrates the steady-state operation of the proposed converter. During phase Φ_A the inductor magnetization period, the switches M_{H1} , M_{H3} , M_{H5} and M_{L2} turn on, the capacitors C_{H1} – C_{H2} get charged while C_{L1} – C_{L2} discharged. During phase Φ_B the inductor demagnetization period, the switches M_{H2} , M_{H4} , M_{H6} and M_{L1} turn on with C_{H1} and C_{H2} directly charge C_{L1} and C_{L2} to the output and constitute a capacitive-current I_C path, in addition to the inductive-current I_L path. For the typical 12V-to-1V DC-DC conversion, by charge and volt-second balance, the converter duty cycle $D = 3V_O / (V_{IN} - 2V_O) = 0.3$ and I_C accounts for 17% of the total load current. Figure 11.8.2 (bottom) presents the voltage swing of each switching node. As is shown, the voltage swing of SW, SW1 and SW3 equals $(V_{IN} - 2V_O)/3 = 3.33V$ while that of SW2, SW4 and SW5 equals $V_O = 1V$. The maximum switch voltage stress is thus 4.3V. Figure 11.8.2 also gives the detailed power rails for drivers of each high-side switch. M_{H1} – M_{H2} employ the stacked boot-strap design from V_{IN} to guarantee a decent drive voltage. For M_{H3} – M_{H5} , the V_{IN} powered boost-strap concept, however, could incur huge driving losses because of the large voltage difference between V_{IN} and the lower stages. Here, instead, each lower switch M_{Hi} is powered by the switching node $SW_{(i-2)}$ considering that a constant floating voltage source of 4.3V exists between each $SW_{(i-2)}$ and SW_i . In this way, the M_{H3} – M_{H5} is driven by a 4.3V floating supply with no boot-strap capacitor required.

Figure 11.8.3 depicts the overall control circuit implementations. Since the proposed work does not require phase shift or capacitor voltage balance control, as other series-capacitor converters usually do, most advanced DC-DC controllers can be easily applied. Here, a ripple-based adaptive-on-time control is implemented for decent transient response. As Fig. 11.8.3 (right) shows, a transconductance stage senses the V_O variations and generates the corresponding on-time to accelerate the transient behavior. Besides, a zero-current-detection (ZCD) circuit is adopted to enhance the light-load efficiency. By

detecting the direction of the SW node during the deadtime period, the optimum off-time is achieved with little power consumption. From the time diagram of Fig. 11.8.3 (bottom), for $I_L > 0$ when the switches turn off, the 7b counter up-counts and extends the off-time. While for $I_L < 0$, vice versa. Finally, the counter settles with the least significant bit toggles and the off-time reaches optimum.

The proposed dual-path series-capacitor converter is fabricated in a 0.18 μ m BCD process for a 9~16V input and 0.6~1.6V output DC-DC conversion with 5A loading capacity. Figure 11.8.4 (top) shows the startup and the steady-state waveforms of each switching nodes for a typical 12V-to-1V design. As is shown, all the switching nodes move up at the same speed while with different voltage swings. During the steady-state, the SW1 and SW3 show 3.33V swing while the SW2, SW4 and SW5 give 1V swing, matching well with the previous analysis of Fig. 11.8.2. Besides, Figure 11.8.4 (bottom) presents the measured inductor current I_L and duty cycle D under different conditions. For $V_{IN} = 12V$ and $I_O = 1A$, the average I_L is 0.73A and 0.83A, and D is 0.55 and 0.33, for V_O equals 1.6V and 1.0V, respectively. For a 12V-to-1V design with $I_O = 0.1A$, the converter operates at discontinuous conduction mode with frequency reduced from 1MHz to 231kHz, leading to light-load efficiency enhancement.

Figure 11.8.5 (top) shows the measured switching node waveforms during the dynamic voltage scaling (DVS) and line transitions. When the output V_O goes from 1V to 1.6V, the swing of SW2 tracks V_O and also changes from 1V to 1.6V, while that of SW1 and SW3 achieve dynamic voltage balance. As V_{IN} steps from 9V to 12V, the SW2 remain the constant $V_O = 1V$ swing while that of SW1 and SW3 self-balanced to higher voltages. Both measurements indicate the strong robustness of the proposed converter. Figure 11.8.5 (bottom) also presents the measured load transitions from 0.5A to 2.5A with 10nS edge time for the 12V-to-1V design. The output V_O shows undershoot and overshoot voltages of $-79mV$ and $100.8mV$, respectively, and the corresponding 0.1% settling time is 11.5 μ s and 12.7 μ s, respectively.

Figure 11.8.6 summarizes the measured efficiency and comparison table. When $V_O = 1V$, the proposed converter achieved 94.2%, 93.4% and 93.0% peak efficiency with V_{IN} of 9V, 12V and 16V, respectively. When $V_{IN} = 12V$, the converter achieved 94.5%, 93.4% and 90.5% peak efficiency with V_O of 1.2V, 1V and 0.6V, respectively. Besides, the converter achieved up to 5A loading capacity with 84.4% efficiency for $I_O = 5A$ and 89.2% efficiency for $I_O = 0.1A$, respectively. The die-area current density is 1.30A/mm². From the measurement results, the prototype demonstrates superior efficiency for the entire voltage conversion range and load ranges. In summary, as compared with the state-of-the-art designs, owing to the reduced voltage stress and current stress by the proposed dual-path series-capacitor design, this work achieves favorable V-A metric and efficiency over wide output voltage and load ranges, even with only one inductor. Besides, inherent full duty cycle operation is supported and controller complexity is greatly reduced.

Figure 11.8.7 shows the die micrograph and a photo of the evaluation board. The die area is 1.96 $\times$ 1.96mm² with the power switches placed around the peripheral to minimize bonding-wire loss and all the bootstrap capacitors are fully integrated.

Acknowledgement:

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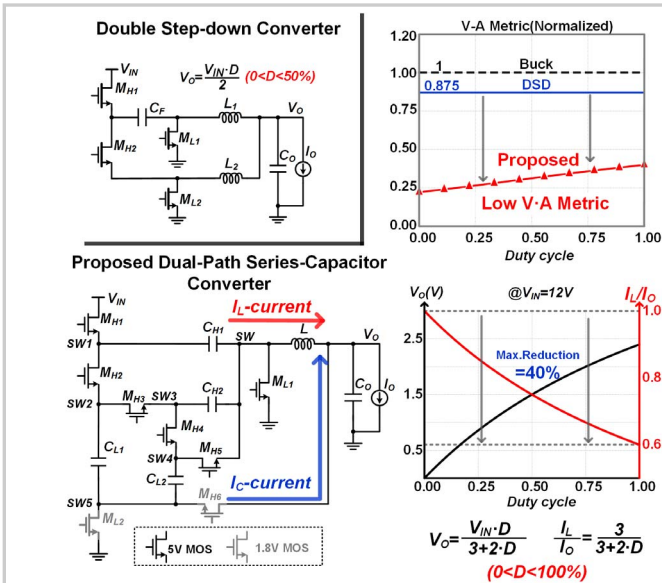


Figure 11.8.1: Double step-down and the proposed dual-path series-capacitor converters.

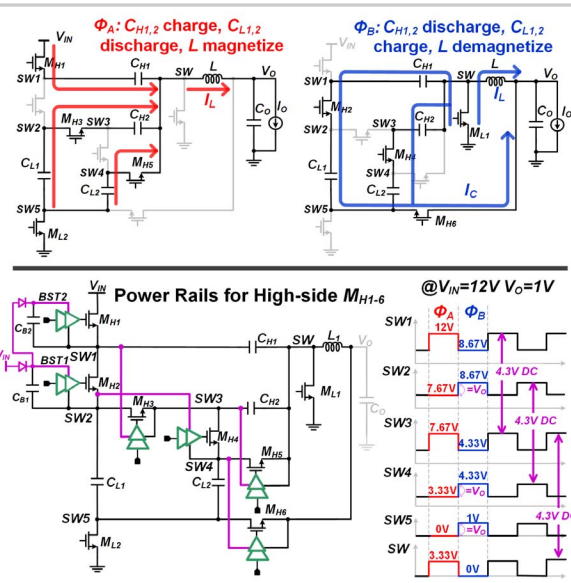


Figure 11.8.2: The steady-state operation principle of the proposed converter and power rails of high-side drivers for MH1-6.

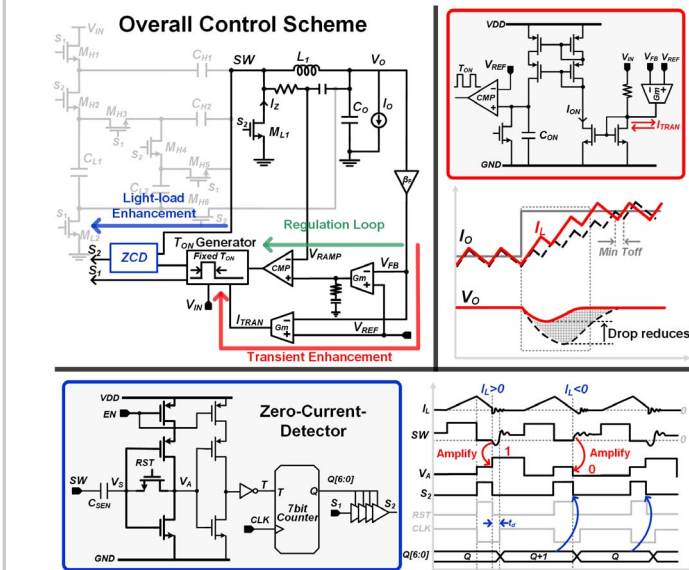


Figure 11.8.3: The overall control scheme of the proposed converter and detailed circuit implementations.

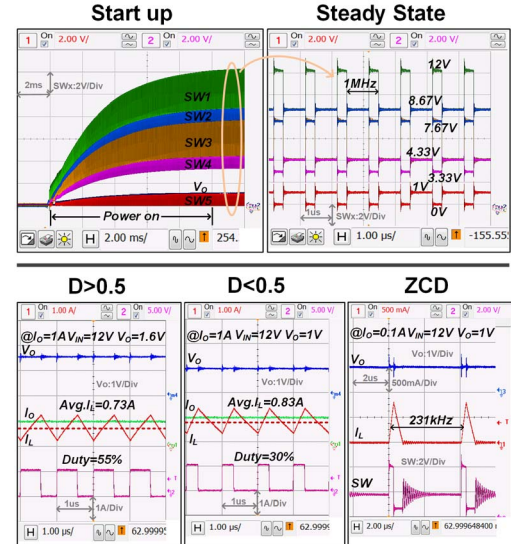


Figure 11.8.4: Measured (top) switching nodes SW and \$V_O\$ waveforms during the power on and steady state operations and (bottom) the inductor current waveforms under different duty cycle and loadings.

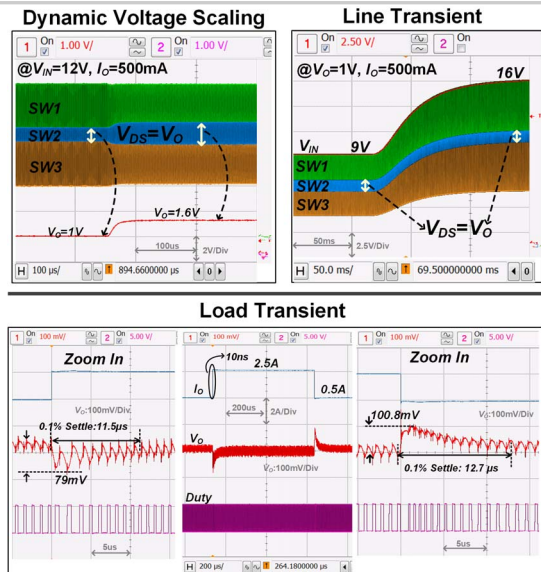
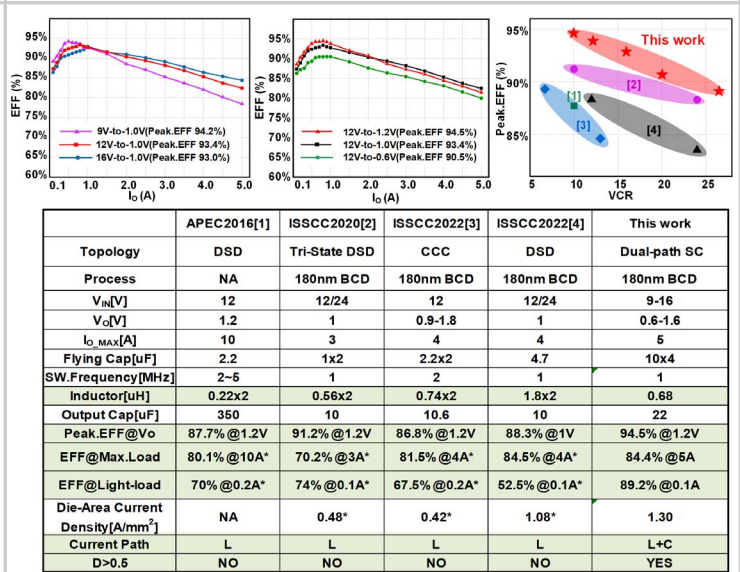


Figure 11.8.5: Measured (top) switching nodes during dynamic voltage scaling and line transient and (bottom) load transient responses between 0.5A and 2.5A.



* Estimated from graph

Figure 11.8.6: Measurement efficiency curves and performance summary.

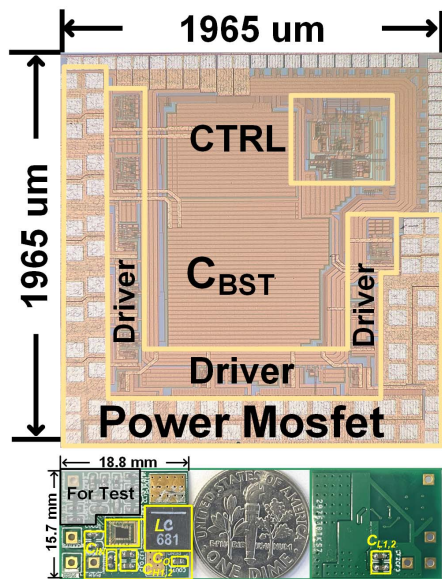


Figure 11.8.7: Chip micrograph and photo of the evaluation board.